INCH-POUND
MIL-PRF-3098/101E
1 November 2017
SUPERSEDING
MIL-PRF-3098/101D
6 March 2013

PERFORMANCE SPECIFICATION SHEET

CRYSTAL UNIT, QUARTZ, CR124/U

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification and MIL-PRF-3098.

Pertinent characteristics: 2.9 MHz to 3.8 MHz; fundamental; noncontrolled; antiresonance.

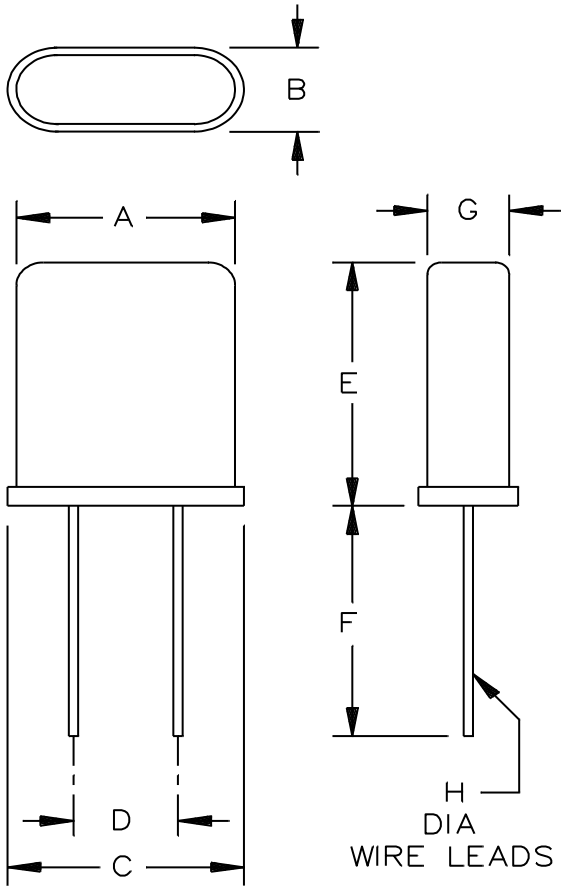


FIGURE 1. Crystal unit - CR124/U.

AMSC N/A
FSC 5955

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Ltr	Inches		mm	
	Min	Max	Min	Max
A	---	.402	---	10.21
B	---	.183	---	4.65
C	---	.435	---	11.05
D	.184	.200	4.67	5.08
E	---	.530	---	13.46
F	.50	---	12.7	---
G	---	.150	---	3.81
H	.015	.019	0.38	0.48

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Marking to be in accordance with MIL-PRF-3098.

FIGURE 1. Type CR124/U, crystal unit - Continued.

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REQUIREMENTS:

Dimensions, marking, and configuration: See figure 1.

Frequency range: 2.9 MHz to 3.8 MHz, inclusive.

Capacitance, shunt: 7 pF, maximum.

Frequency tolerance:

Operating temperature range: ±50 ppm.

Mode of oscillation: Fundamental.

Rated drive level: 1.0 mW, maximum.

Calibration values:

Resistance: 150 ohms.

Crystal current: 2.7 mA.

Antiresonance, load capacitance: 32.0 pF ±.05 pF.

Operating temperature range (noncontrolled): -55°C to +100°C, inclusive.

Operable temperature range: -65°C to -55°C and +100°C to +105°C, inclusive.

Equivalent resistance: 450 ohms, maximum.

Shock:

Frequency change permitted: ±5 ppm.

Equivalent resistance change permitted: ±15 percent.

Vibration: MIL-STD-202-204, test condition A; 0.10 inch double amplitude or 5 gravity units (55 cps to 500 cps), whichever is less.

Frequency change permitted: ±5 ppm.

Equivalent resistance change permitted: ±15 percent.

Thermal shock: ±50 ppm.

Aging:

Frequency change permitted: ±5 ppm.

Reference documents. In addition to MIL-PRF-3098, this document references the following:

MIL-STD-202-204

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Review activities:
Air Force - 19, 84, 99

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(Project 5955-2017-011)

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